Satpreet Makhija

satpreet.makhija@ashoka.edu.in
satpreetmakhija.com

Research Statement

My research interests lie in programming-language semantics, especially probabilistic programming, and in causal inference, type systems, and formal verification. I study how formal semantics makes programs and causal models precise, how type systems enforce invariants, and how verification establishes that implementations satisfy their specifications. My broader goal is to develop rigorous foundations for reliable computational systems. Before moving into research, I worked on reliable, air-gapped software systems at Google Cloud.

Publications

Graph Surgery and the Do-Operator: A Precise Correspondence for Acyclic Structural Causal Models
Satpreet Makhija. *arXiv:2608.17634 [cs.AI]*, Aug 2026. arXiv · PDF · Lean 4 formalization

Education

Ashoka University — Sonipat, Haryana
Bachelor of Science (Honours) in Computer Science — *Aug 2018 - Apr 2022*

Major CGPA: 3.74/4

Relevant Coursework

Discrete Mathematics; Algorithm Design and Analysis; Linear Algebra; Calculus; Abstract Algebra; Probability Theory; Programming Language Design and Implementation; Operating Systems; Computer Networks; Computer Security and Privacy.

Summer Schools and Workshops

- Indian School on Logic and its Applications (ISLA 2026) · Krea University · Jun 25–30, 2026. Courses: Programs and Proofs; Computability Theory.
- Formal Methods Update Meeting 2026 · Krea University · Jul 2–3, 2026.

Research and Professional Experience

Ashoka University — Sonipat, Haryana
Research Associate · Programming Languages — *Jun 2026 - Present*

- Studying semantics of programming languages, computational effects, and categorical approaches to programming-language foundations.
- Mechanized the paper's principal results in Lean 4, including graph-mechanism correspondence, sequential interventions, and ancestor locality, in a reproducible, CI-checked artifact with a pinned toolchain and axiom audit.

Firebolt — Bangalore, India
Software Engineer · Self-hosted Platform — *Nov 2025 - Mar 2026*

- Joined the founding engineering team building Firebolt's first self-hosted platform for isolated, air-gapped enterprise deployments, including feature flags and Temporal workflow orchestration.

Google — Bangalore, India
Software Engineer · Google Cloud, Distributed Cloud Team — *Mar 2024 - Sep 2025*

- Built software for Google Distributed Cloud, bringing Google Cloud services into air-gapped, sovereign, and regulated environments through distributed systems that operated without public-network dependencies.

- Designed a Harbor-based artifact registry for a 40kg portable data center, adding Kubernetes self-healing, node-local fallback mirrors, and image signature verification to support offline bootstrap and operation.
- Built APIs and observability tooling for Prometheus/Grafana monitoring, covering system health, storage, image-pull latency, error rates, alerting, and dashboards for internal teams and deployment operators.
- Mentored junior engineers and hosted a summer intern who later joined full time.

Google Bangalore, India
Strategic Cloud Engineer · Google Cloud *Jul 2022 - Mar 2024*

- Helped enterprises become cloud-native on Google Cloud, combining technical consulting, hands-on implementation, and architecture guidance across cloud infrastructure projects.
- Researched and prototyped Data Insights, an LLM-powered natural-language layer for tabular data discovery, testing generated questions, SQL guardrails, accuracy, and latency for cold-start exploration of unfamiliar datasets; the capability later became part of a Google Cloud data management product.
- Built reusable cloud and data tooling, including Terraform modules for Kubernetes clusters and VPCs alongside open-source Dataproc templates for Spark/Hadoop ETL.

Teaching

Ashoka University Sonipat, Haryana
Teaching Fellow · Data Structures and Algorithms (CS-2212-1) *Monsoon 2026*

- Teaching with Prof. Aalok Thakkar in the Monsoon 2026 offering of the course.

AshokaX Online
Teaching Assistant · Machine Learning *Apr 2022 - Jun 2022*

- Assisted Prof. Debayan Gupta in teaching machine learning to high school students.

Ashoka University Sonipat, Haryana
Teaching Assistant · Computer Organization and Systems *Monsoon 2021*

- Assisted Prof. Manu Awasthi with Computer Organization and Systems in Ashoka's Computer Science department.